

2018

**COMPUTER SCIENCE – GENERAL**

**Second Paper**

**Group - C**

**(Database Management System)**

7. (a) What are data redundancy and data inconsistency? Explain with example. 4  
(b) Distinguish between strong and weak entity with an example. 3  
(c) Discuss utility of 'Foreign key' and 'composite primary key' with example. 6  
(d) Define 3-NF. 3

8. (a) Consider the following relational schema :

EMPLOYEE (*EID*, EName, Date of Birth, Salary, DNo)

DEPARTMENT (*DNo*, DName, Location)

Write SQL commands for the following queries :

- (i) Find all employees whose names start with 'A'.  
(ii) Find all employees of 'Marketing' department.  
(iii) Find all departments located in Kolkata.  
(iv) Find all employees who earn salary between ₹ 30,000 to ₹ 80,000. 3+3+3+3  
(b) Explain super key and primary key with examples. 4
9. Write short notes on *any four* : 4×4
- (a) Loss less decomposition  
(b) Aggregate functions in SQL  
(c) ANSI/SPARC architecture  
(d) File organization  
(e) Role of DBA  
(f) Network data model